Safa Berberoğlu

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Profile

Dynamic 3rd-year Computer Engineering student with a strong foundation in scalable backend architectures, cloud computing, and IoT integrations. Passionate about bridging the gap between hardware and software to create innovative smart home solutions and connected devices. Seeking to leverage my expertise in Python, AWS, and embedded systems to contribute to high-impact automation systems and data-driven IoT architectures at a leading home appliance technology company.

Professional Experience

Founding Vice President & Technical Lead, *AWS Cloud Club at Trakya University*
3/2026 – Present
Edirne, Turkey

- Co-founded the university's first official AWS Cloud Club, leading the organizational setup, core team recruitment, and strategic planning from the ground up.
- Spearheaded initial outreach and promotional campaigns to introduce AWS Cloud technologies and club objectives to the engineering student body.

Technical Lead, Huawei *Student Developers at Trakya University*
10/2025 – 3/2026
Edirne, Turkey

- Spearheaded hands-on hardware bootcamps, training an 80-person cohort in embedded systems and IoT architecture from scratch.

IT Support Specialist (Part-Time), *Trakya University, Department of Information Technology*
12/2025 – Present
Edirne, Turkey

- Managed database operations and utilized graphical user interfaces (GUI) to ensure smooth technical workflows.
- Provided technical support for mobile applications and managed operational data tracking workflows to ensure accurate reporting.

IT & Administrative Assistant (Part-Time), *Trakya University, Department of Student Affairs*
01/2025 – 06/2025
Edirne, Turkey

- Streamlined student form processing and document management operations using MS Office applications.
- Designed and deployed the English version of the Student Affairs website to improve accessibility and user experience for international students.

Education

Computer Engineering, *Trakya University*
10/2023 – 06/2027
Edirne, Turkey

- Relevant Coursework: Software Engineering, Data Structures, Web Technologies, and Embedded Systems.
- Activities: Developing hardware-software integrated projects, organizing technical workshops, and actively participating in developer communities.

Areas of Expertise

Backend & Cloud
- .NET/ASP.NET Core
- Java
- Python
- FastAPI
- AWS

Frontend & Mobile
- Flutter
- Dart
- HTML/ CSS/ JavaScript

Database
- SQL
- PostgreSQL
- Firebase

Core Engineering
- C/ C++
- Embedded Systems
- IoT

Soft Skills
- Teamworking
- Leadership
- Problem Solving
- Creativity

Languages
- English C1

Projects

DevDiary, *Privacy-Focused AI Note Analyzer*
2026

- Engineered a privacy-centric backend utilizing FastAPI to integrate Local LLMs for automated note categorization and sentiment analysis.
- Developed a secure RESTful API architecture with PostgreSQL to manage data persistence while ensuring zero-cloud dependency for user data.

AWS Racer, *Cross-device synchronization via low-latency cloud messaging and serverless architecture.*
2026

- Engineered a high-performance communication bridge using Cloud-based MQTT/ WebSockets to achieve sub-100ms latency between mobile controllers and display engines.
- Implemented a Real-time State Synchronization (Shadowing) architecture to manage seamless device interaction and digital-twin consistency across the network.
- Developed a scalable backend infrastructure utilizing NoSQL cloud storage for global leaderboards and serverless logic for session management.

Smart Remote, *IoT TV Control System*
2026

- Engineered a real-time IoT system integrating NodeMCU (C++) and a Flutter application via MQTT protocol for low-latency device control and state synchronization.
- Implemented an IR-based hardware interface to bridge physical home appliances with a modern mobile dashboard.

ASA, *Solar Storms Early Warning System (TUA Astro Hackathon)*
2026

- Architected a FastAPI-based early warning platform that processes real-time NASA/NOAA data and utilizes a Scikit-learn ML model to predict geomagnetic disturbances (Kp-index).
- Engineered a low-latency data pipeline via WebSockets and implemented L1 point delay analysis for 72-hour proactive forecasting with automated Telegram alerts.